Integration by Substitution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: u-substitution

Evaluate each integral using substitution:

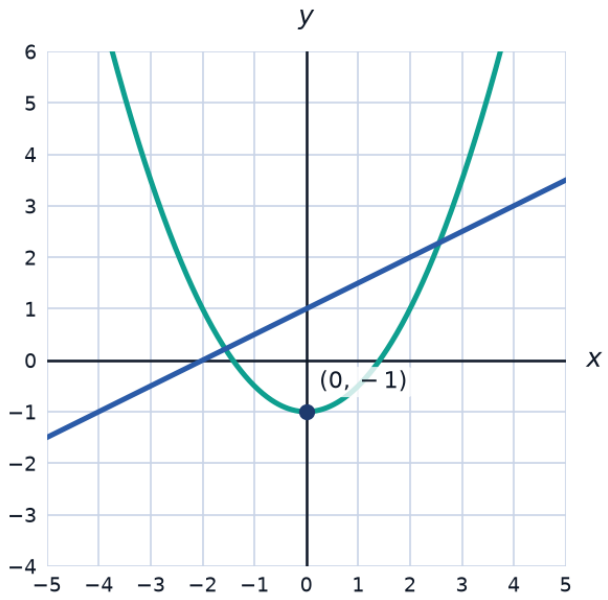
a $\int 2x(x^2+1)^5\,dx$

b $\int \frac{x}{\sqrt{x^2+9}}\,dx$

c $\int \cos(3x)\,dx$

2 [Foundational] • Target: u-substitution

Evaluate $\int x^2 e^{x^3}\,dx$.

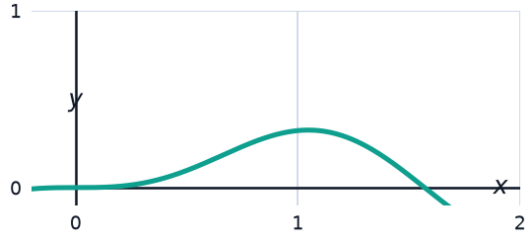


3 [Foundational] • Target: Definite integrals with substitution

Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly, changing the limits with $u = x^2 + 1$.

4 [Foundational] • Target: Definite integrals with substitution

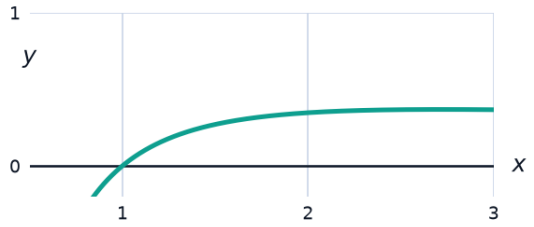
Evaluate $\int_0^{\pi/2} \sin^3 x \cos x\,dx$.



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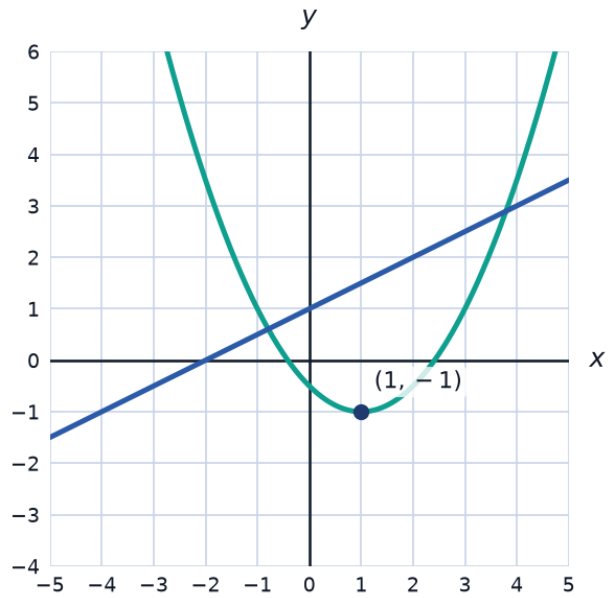
5 [Foundational] • Target: Definite integrals with substitution

Evaluate $\int_1^e \frac{\ln x}{x}\,dx$.



6 [Foundational] • Target: Substitution with trigonometric and exponential integrands

Evaluate $\int \frac{2x+3}{x^2+3x+1}\,dx$, recognizing the numerator as the derivative of the denominator.



7 [Foundational] • Target: Choosing the substitution

For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3-1)^7\,dx$

b $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

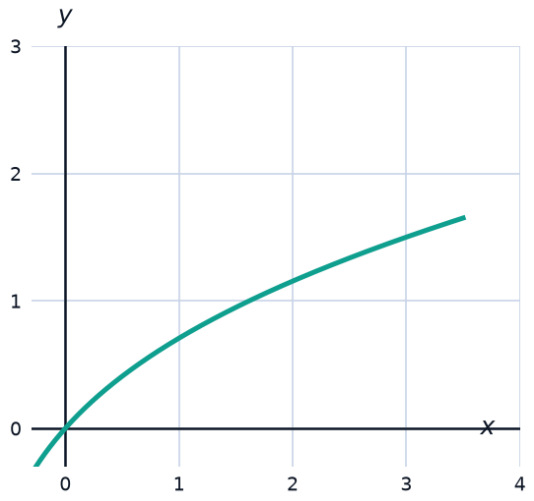
8 [Intermediate] • Target: Substitution with radicals

Evaluate $\int x\sqrt{2-x}\,dx$ using the substitution $u = 2 - x$ (so $x = 2 - u$).

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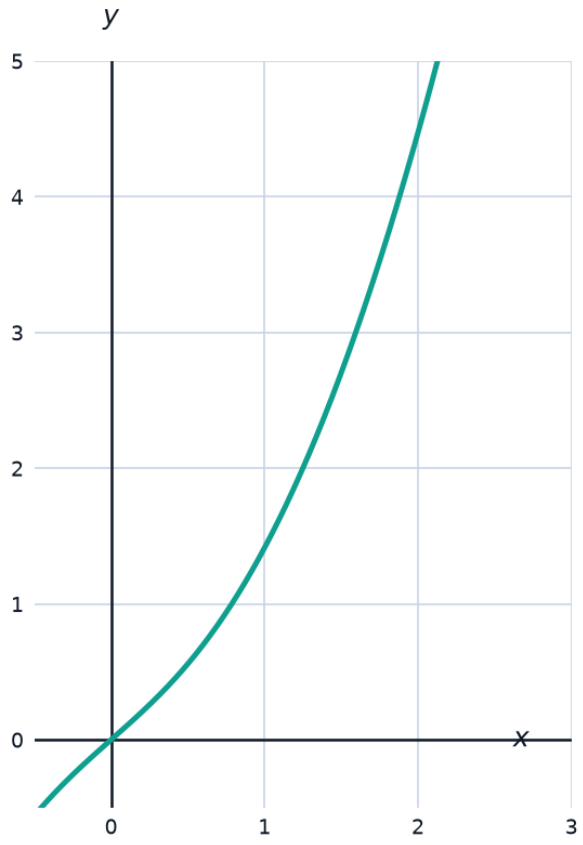
9 [Intermediate] • Target: Substitution with radicals

Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$ using $u = x + 1$ (so $x = u - 1$), changing the limits.



10 [Intermediate] • Target: Visualizing an area found by substitution

The graph shows $f(x) = x\sqrt{x^2+1}$ on $[0, 2]$. Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly using $u = x^2 + 1$, and match it to the shaded area.



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11 [Intermediate] • Target: Substitution with double angle and reciprocal forms

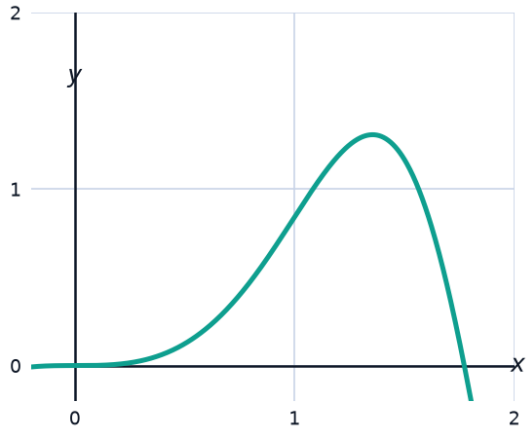
Evaluate $\int \sin x \cos x \, dx$ two ways: (a) using $u = \sin x$, and (b) using the identity $\sin x \cos x = \frac{1}{2}\sin(2x)$. Confirm both give equivalent answers.

12 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \frac{1}{x \ln x} \, dx$ using $u = \ln x$.

13 [Intermediate] • Target: Substitution in motion and rate problems

A particle's velocity is $v(t) = t\sin(t^2)$ m/s. Find its displacement from $t = 0$ to $t = \sqrt{\pi}$, using $u = t^2$.



14 [Intermediate] • Target: Substitution in motion and rate problems

Evaluate $\int \frac{(\ln x)^2}{x} \, dx$ using $u = \ln x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

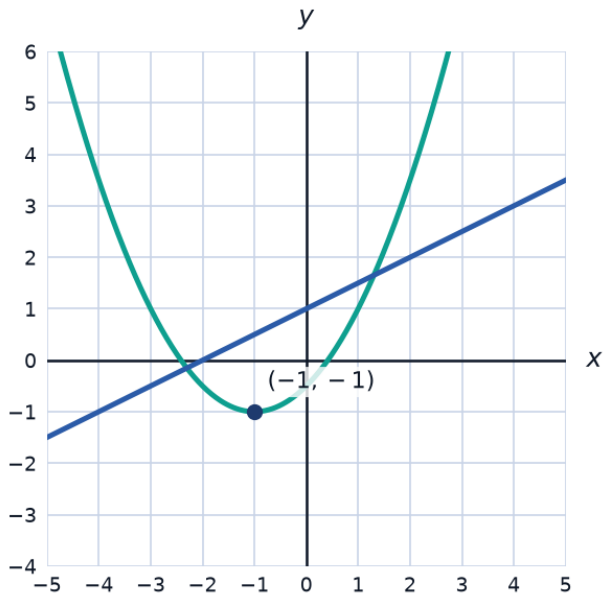
15 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 2.01 Evaluating Limits

Evaluate $\int (x+1)\sqrt{x^2+2x+5} \, dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.

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16 [Challenging] • Target: Substitution with trigonometric and exponential integrands • Cross-Lesson: 2.02 Continuity

Net change in population from $t = 0$ to $t = T$ with rate $r(t)$:.



17 [Challenging] • Target: Choosing the substitution • Cross-Lesson: 3.01 The Derivative from First Principles

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:.

18 [Challenging] • Target: Substitution with radicals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

At what rate is the area of a square changing when its side is 4 m and growing at 2 m/s? Show your complete algebraic work.

19 [Challenging] • Target: Visualizing an area found by substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Under the change of variables $x = 2u - v$, $y = 2u + 4v$, a region S in the uv-plane with area 6 maps to a region R. Find the area of R.

20 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Under the change of variables $x = 2u$, $y = -2u + 4v$, a region S in the uv-plane with area 2 maps to a region R. Find the area of R.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic u-Substitution

Evaluate $\int 2x(x^2+5)^4\,dx$.

a. $\frac{1}{5}(x^2+5)^5+C$

b. $\frac{1}{4}(x^2+5)^4+C$

c. $(x^2+5)^5+C$

d. $\frac{2}{5}(x^2+5)^5+C$

22 [Foundational] • Target: Basic u-Substitution

Evaluate $\int x^2\sqrt{x^3+1}\,dx$.

a. $\frac{2}{9}(x^3+1)^{3/2}+C$

b. $\frac{2}{3}(x^3+1)^{3/2}+C$

c. $\frac{1}{3}(x^3+1)^{3/2}+C$

d. $\frac{3}{2}(x^3+1)^{1/2}+C$

23 [Foundational] • Target: Basic u-Substitution

Evaluate $\int \frac{x}{x^2+9}\,dx$.

a. $\frac{1}{2}\ln(x^2+9)+C$

b. $\ln(x^2+9)+C$

c. $\frac{1}{3}\arctan(x/3)+C$

d. $\frac{1}{2(x^2+9)^2}+C$

24 [Foundational] • Target: Trigonometric and Exponential Substitutions

Evaluate $\int \sin^3(x)\cos(x)\,dx$.

a. $\frac{1}{4}\sin^4(x)+C$

b. $\frac{1}{3}\sin^3(x)+C$

c. $-\frac{1}{4}\cos^4(x)+C$

d. $\sin^4(x)+C$

25 [Foundational] • Target: Trigonometric and Exponential Substitutions

Evaluate $\int \tan(x)\,dx$.

a. $\ln|\sec x|+C$

b. $-\ln|\sec x|+C$

c. $\sec^2(x)+C$

d. $\ln|\sin x|+C$

26 [Foundational] • Target: Trigonometric and Exponential Substitutions

Evaluate $\int e^{3x-1}\,dx$.

a. $\frac{1}{3}e^{3x-1}+C$

b. $3e^{3x-1}+C$

c. $e^{3x-1}+C$

d. $\frac{1}{3}e^{3x}+C$

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27 [Foundational] • Target: Definite Integrals with Limits Change

Evaluate $\int_0^1 x(x^2+1)^3\,dx$.

a. 15/8

b. $16/8 = 2$

c. 7/8

d. 15/4

28 [Foundational] • Target: Definite Integrals with Limits Change

Evaluate $\int_0^{\sqrt{\pi}} x\sin(x^2)\,dx$.

a. 1

b. 0

c. 2

d. 1/2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Advanced Linear Substitutions

Evaluate $\int x\sqrt{x+2}\,dx$.

a. $\frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$

b. $\frac{2}{3}(x+2)^{3/2} + C$

c. $\frac{1}{5}(x+2)^{5/2} - \frac{2}{3}(x+2)^{3/2} + C$

d. $\frac{2}{5}x^{5/2} + C$

30 [Intermediate] • Target: Advanced Linear Substitutions

Evaluate $\int \frac{x}{\sqrt{x+1}}\,dx$.

a. $\frac{2}{3}(x+1)^{3/2} - 2\sqrt{x+1} + C$

b. $\frac{1}{3}(x+1)^{3/2} - \sqrt{x+1} + C$

c. $2\sqrt{x+1} + C$

d. $\frac{2}{3}(x+1)^{3/2} + C$

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31 [Intermediate] • Target: Advanced Linear Substitutions

Evaluate $\int \frac{e^{2x}}{e^x + 1}\,dx$.

a. $e^x - \ln(e^x + 1) + C$

b. $e^x + \ln(e^x + 1) + C$

c. $\ln(e^x + 1) + C$

d. $\frac{1}{2}e^{2x} + C$

32 [Intermediate] • Target: Definite Integrals with Limits Change

Under the change of variables $x = 3u + v$, $y = 3u + 3v$, a region S in the uv-plane with area 3 maps to a region R. Find the area of R.

a. 3

b. 6

c. 18

d. $\frac{1}{2}$

33 [Intermediate] • Target: Advanced Linear Substitutions

Under the change of variables $x = 4u + v$, $y = 3u + 3v$, a region S in the uv-plane with area 4 maps to a region R. Find the area of R.

a. 36

b. 4

c. 9

d. 28

34 [Intermediate] • Target: u-substitution

Under the change of variables $x = 3u + 2v$, $y = u + 2v$, a region S in the uv-plane with area 5 maps to a region R. Find the area of R.

a. 20

b. 5

c. 4

d. 25

35 [Intermediate] • Target: Definite integrals with substitution

Under the change of variables $x = 4u + v$, $y = -u + v$, a region S in the uv-plane with area 6 maps to a region R. Find the area of R.

a. 6

b. 5

c. 30

d. $\frac{6}{5}$

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36 [Intermediate] • Target: Substitution with trigonometric and exponential integrands

Under the change of variables $x = 3u - 2v$, $y = -u + 2v$, a region S in the uv-plane with area 4 maps to a region R. Find the area of R.

a. 4

b. 20

c. 1

d. 16

37 [Intermediate] • Target: Choosing the substitution

Use a double integral to find the area of the region under $y = x^2$ for $0 \le x \le 1$.

a. $\frac{2}{3}$

b. $\frac{1}{3}$

c. $\frac{1}{6}$

d. $\frac{1}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Substitution with radicals • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Weierstrass substitution $t = \tan(x/2)$: $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$. Converts rational trig integrands to:

a. trig of t

b. rational functions of t

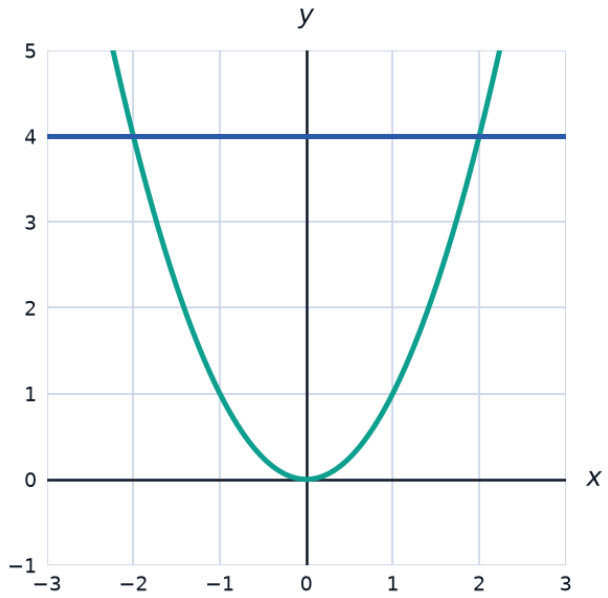
c. exponentials of t

d. polynomials

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39 [Challenging] • Target: Visualizing an area found by substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Consider the region R in the xy-plane bounded by the curves $y = x^2$ and $y = 4$. What is the area of this region?



a. $\frac{32}{3}$

b. $\frac{16}{3}$

c. 8

d. 16

40 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Evaluate $\int_0^2 \int_y^2 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{1}{2} + \frac{e^4}{2}$

b. $-1 + e^4$

c. $\frac{1}{2} + \frac{e^4}{2}$

d. $-\frac{1}{4} + \frac{e^4}{4}$

41 [Challenging] • Target: Substitution in motion and rate problems • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Evaluate $\int_0^{\sqrt{\pi}} \int_x^{\sqrt{\pi}} \sin(y^2)\,dy\,dx$ by reversing the order of integration.

a. 1

b. 2

c. $\frac{1}{2}$

d. 3

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42 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 6.01 Related Rates

A ladder 13 m long leans against a wall. Base slides at 2 m/s. Rate of top falling when base is 5 m from wall:

a. $-\frac{5}{6}$ m/s

b. $-\frac{5}{12}$ m/s

c. $-\frac{2}{5}$ m/s

d. $-\frac{1}{2}$ m/s

43 [Challenging] • Target: Basic u-Substitution • Cross-Lesson: 6.02 Optimization and Curve Sketching

Region $x \geq 0$, $y \geq 0$, $x + y \leq 1$: volume when rotated about y-axis by shell method:

a. $\frac{\pi}{3}$

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{6}$

44 [Challenging] • Target: Trigonometric and Exponential Substitutions • Cross-Lesson: 7.02 Integration by Parts

Area below x-axis from -2 to 0 of $f(x) = x^2 - 4$:

a. $-\frac{16}{3}$

b. $\frac{8}{3}$

c. $\frac{16}{3}$

d. 4

45 [Challenging] • Target: Definite Integrals with Limits Change • Cross-Lesson: 8.01 Area Between Curves

Washer method: volume when $R(x)$ is outer and $r(x)$ inner radius:

a. $\pi\int (R - r)^2 dx$

b. $\pi\int (R^2 - r^2) dx$

c. $2\pi\int (R - r) dx$

d. $\pi\int R^2 dx - \pi\int r^2 dx$ (same)

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